ECOLE SUPERIEURE DE GESTION, D'INFORMATIQUE ET DES SCIENCES
ACADEMIC YEAR: 2022-2023

Summative Evaluation/ IRT1 **Duration: 2H00** **Examiner: Dr. Edouard Koba**

COMPUTER SCIENCE ENGLISH

I- READING COMPREHENSION

Text: From mainframes to microcomputers

1. The microprocessor: At the heart of a microcomputer is a microprocessor, a simple integrated circuit that contains an arithmetic/logic unit as well as control capability for memory and input/output access. The microprocessor controls the sequence of operations and the arithmetic and logical operations, as storing data, instructions and intermediate and final results, much the same as CPU of mainframe computer. The CPU of a mainframe computer, however, is composed of a series of integrated circuits and is much more sophisticated than the microprocessor. Although miniature in size, the microprocessor is, in some microcomputers, capable of performing up to 700,000 additions per second. There are a number of differences between mainframe computers and microcomputers; however, the prefix micro should be interpreted in terms of size and rather than capability.

2. Microcomputers versus Mainframes: As indicated, the microprocessor and mainframe's CPU perform nearly identical function, yet the manner in which each carries out these functions differs greatly. The CPU of a mainframe computer has a much larger and more powerful instruction set to choose from when performing its functions. Microcomputers, for example, may take several processing steps to perform a function performed in one processing step by a mainframe computer. The instruction set is designed into the electronic circuitry of the computer by engineers. The factors that affect the size of the instruction set are the register size and word size. The registers are the locations within the CPU that are used as temporary storage areas to facilitate the transfer of data and instructions. Word size is based on the size of the registers, and it is the word size that ultimately affects the instruction set designed into number of instructions in the instruction set.

3. The IBM 370/3033 mainframe computer, for example, has a word size of 64 bits, while the APPLE microcomputer has an 8-bit word size, and the IBM personal computer has 16-bit word size. The architectural design of the mainframe, therefore, permits a larger instruction set, access to more memory in a single step, and greater speed in processing large number - arithmetic computations. Also since microcomputers have smaller register sizes for holding data, they perform arithmetic computations with less precision. The number of registers is significant factor to consider when comparing microcomputers to mainframe computers. Microcomputers also have fewer registers and because of this they are considerably slower than mainframe computers when performing arithmetic computations.

4. Another difference between microcomputer and mainframe computers is that mainframes allow for multiple users, while most microcomputers allow for only one user at time. Microcomputers are limited to one user because one communication channel is used to

communicate with the computer's peripheral devices. For this reason, the software required to run the computer (the operating system) is significantly less complex. Mainframes, on the other hand, use a concept called parallelism to allow for what appears to be simultaneous data movement through a number of communication channels. Mainframes are designed to accept a maximum number of peripheral devices to each channel.

QUESTIONS

A. ***Answer by R (= right) or W (=wrong) according to the reading text.*** **(***16 points***)**

1. The microprocessor is a simple integrated circuit while the mainframe's CPU is made up of series of integrated circuits.
2. The prefix micro should either be interpreted in terms of size or in terms of capability.
3. The microprocessor and mainframe's CPU perform identical functions.
4. The manners in which the microprocessor and mainframe's CPU carry out their functions are nearly identical.
5. Registers are parts beside the CPU that are used as transitory storage areas.
6. Microcomputers perform arithmetic computations with more precision.
7. Mainframe computers are considerably faster than microcomputers when performing arithmetic computations on account of having fewer registers.
8. While most microcomputers allow for at most more than one user at time, mainframe computers allow for many users at time.

B. ***Answer these questions with information from the text.*** **(***16 points***)**

1. How do you understand the prefix "micro" in the word microcomputer and where do you locate the microprocessor in the system unit?
2. How do you account for the common and opposite points between the microcomputer's microprocessor and mainframe's CPU?
3. What are the likenesses and dissimilarities between the microcomputer and mainframe?
4. What differentiates the IBM 370/3033 mainframe computer from the APPLE microcomputer and IBM personal computer?

C. Translate the following passage into French. **(***8 points***)**

"Another difference between microcomputer and mainframe computers is that mainframes allow for multiple users, while most microcomputers allow for only one user at time. Microcomputers are limited to one user because one communication channel is used to communicate with the computer's peripheral devices. The software required to run the computer (the operating system) is significantly less complex. Mainframes use a concept called parallelism to allow for what appears to be simultaneous data movement through a number of communication channels. "

II LINGUISTIC PERFORMANCE

A. Choose the best adjective. Write down just the number and letter. (*10 points*)

1. Oh dear. I pressed the _____ button.
a. incorrect **b.** wrong **c.** false
2. I can't use my mobile phone. The battery's _____.
a. over **b.** flat **c.** exhausted
3. The battery isn't completely flat, but it's very _____.
a. down **b.** short **c.** low
4. My video camera is very _____.
a. easy to use **b.** uncomplicated **c.** obvious
5. My new computer has a very _____ processor.
a. quick **b.** high speed **c.** fast
6. The X19 notebook computer features a very _____ design.
a. compact **b.** little **c.** small
7. Keeping files on paper is _____ solution.
a. an old-tech **b.** a past-tech **c.** a low-tech
8. Keeping files on a computer database is a _____ solution.
a. new-tech **b.** now-tech **c.** high-tech
9. My new PDA is the _____ model.
a. latest **b.** newest **c.** most modern
10. In our office, we've set up a _____ network.
a. wire-free **b.** no wires **c.** wireless

B. *Choose the best verb. Write down just the number and letter.* (*10 points*)

1. To turn on the computer, _____ the "Start" button.
a. touch **b.** press **c.** switch
2. The printer has _____ of ink.
a. finished **b.** ended **c.** run out
3. Unfortunately, my scanner isn't _____ at the moment.
a. working **b.** going **c.** doing
4. Please _____ the CD ROM.
a. insert **b.** introduce **c.** inject
5. The projector isn't working because it isn't _____.
a. plugged **b.** plugged in **c.** plugged into
6. The batteries in my digital camera are nearly dead. They need _____.
a. to change **b.** exchanging **c.** changing
7. I have to _____ a computer screen for eight hours a day.
a. see **b.** look at **c.** watch
8. Switch off your computer, and _____ it from the wall socket.
a. de-plug **b.** unplug **c.** non-plug

9. I turned off the photocopier and _____ the plug.
a. pulled out **b.** extracted **c.** took away
10. _____ any key to continue.
a. Kick **b.** Smash **c.** Hit

C. ***Use the correct form or tense of the bracketed verbs. Write down just the number and the correct verb form and tense.*** **(***10 points***)**

1. It is long time since this power plant **(to sell)** to westerners.
2. If John **(to know),** he would not have married this unbearable and *faro-faro* wife.
3. These policeman **(to regulate)** the traffic for four hours now.
4. The new elected president has done nothing for about three months but **(to abrogate)** decrees of the former regime.
5. You really confirm that cars **(not to make)** in your country.

D. ***Choose the most appropriate option from the bracket. Write down just the number and your selection.*** **(***10 points***)**

1. Most students would rather the Head of Studies (**wouldn't resign; didn't resign**) next year.
2. They asked whether I was the man who had called two days (**ago; before**)
3. Most people believed that the chairman was a very wonderful friend of (**us; ours**).
4. The teacher often stops (**to pick up; picking up**) the cell phone when he gets an income call.
5. Unfortunately, I did not remember (**to take; taking**) the bag last time
6. The first problem was the easiest, so (**anybody; nobody)** could do it quickly.
7. In a free service restaurant, everyone serves (**themselves; himself**) after paying the bill.
8. My people sent me (**a; an**) SMS but I was in an area where there was no network.
9. These carriers require that he (**signs; sign**) the delivery note as a proof of goods reception.
10. How (**much; heavy**) did the new born weigh? The pediatrician asked.

III <u>COMPOSITION</u> **(***20 points***)**

Write a-twenty-line paragraph ***(maximum)*** in which you discuss the major reasons which motivated you to enroll/register in ESGIS University as a computer science student.